```
6CSEC2@debian: ~/Desktop/230905248/lab1
File  Edit  View  Search  Terminal  Help
6CSEC2@debian:~/Desktop/230905248/lab1$ mpicc q1.c -o q1 -lm
6CSEC2@debian:~/Desktop/230905248/lab1$ mpirun -np 5 ./q1
Rank 0: 2^0 = 1
Rank 2: 2^2 = 4
Rank 3: 2^3 = 8
Rank 4: 2^4 = 16
Rank 1: 2^1 = 2
6CSEC2@debian:~/Desktop/230905248/lab1$
```

```
6CSEC2@debian: ~/Desktop/230905248/lab1
File  Edit  View  Search  Terminal  Help
6CSEC2@debian:~/Desktop/230905248/lab1$ mpicc q1-2.c -o q1-2
6CSEC2@debian:~/Desktop/230905248/lab1$ mpirun -np 5 ./q1-2
Rank 0: Hello
Rank 1: World
Rank 3: World
Rank 2: Hello
Rank 4: Hello
6CSEC2@debian:~/Desktop/230905248/lab1$
```

```
6CSEC2@debian: ~/Desktop/230905248/lab1
File  Edit  View  Search  Terminal  Help
6CSEC2@debian:~/Desktop/230905248/lab1$ mpicc calc.c -o calc
6CSEC2@debian:~/Desktop/230905248/lab1$ mpirun -np 4 ./calc
Division: 20 / 10 = 2
Subtraction: 20 - 10 = 10
Multiplication: 20 * 10 = 200
Addition: 20 + 10 = 30
6CSEC2@debian:~/Desktop/230905248/lab1$
```

```
6CSEC2@debian: ~/Desktop/230905248/lab1
File  Edit  View  Search  Terminal  Help
6CSEC2@debian:~/Desktop/230905248/lab1$ mpicc q3.c -o q3
6CSEC2@debian:~/Desktop/230905248/lab1$ mpirun -np 5 ./q3
Rank 2: l
Rank 3: l
Rank 4: o
Rank 0: h
Rank 1: e
6CSEC2@debian:~/Desktop/230905248/lab1$
```

```
6CSEC2@debian: ~/Desktop/230905248/lab1
File  Edit  View  Search  Terminal  Help
6CSEC2@debian:~/Desktop/230905248/lab1$ mpicc q4.c -o q4
6CSEC2@debian:~/Desktop/230905248/lab1$ mpirun -np 5 ./q4
Rank 2 Factorial = 2
Rank 3 Fibonacci = 2
Rank 4 Factorial = 24
Rank 0 Factorial = 1
Rank 1 Fibonacci = 1
6CSEC2@debian:~/Desktop/230905248/lab1$
```

```
6CSEC2@debian: ~/Desktop/230905248/lab1
File  Edit  View  Search  Terminal  Help
6CSEC2@debian:/Desktop/230905248/lab1$ mpirun -n 9 exe/q5
[3] 4321
[5] 41
[8] 8291
[6] 801
[7] 51
[4] 103
[0] 81
[1] 325
6CSEC2@debian:/Desktop/230905248/lab1$ mpirun -n 2 exe/q6
[1] 53 59 61 67 71 73 79 83 89 89 97
[0] 2 3 5 7 11 13 17 19 23 29 31 37 41 44 43 47
```